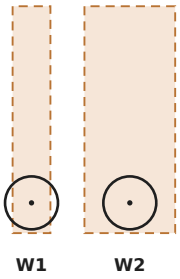
# T4 Book interconnects



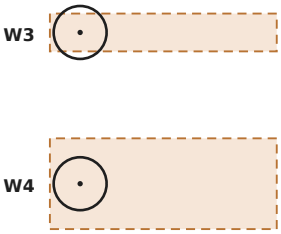
Print sheets 1 and 2 on the two sides of one card, flipping on the long edge. Make three.

V2 bench test

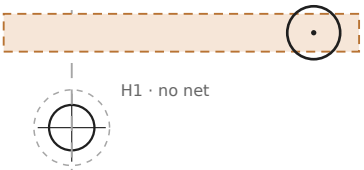



## 1 Edge wraps

Lay each W lane on side A, run the tape over the edge and down side B to the test point behind. W1 and W2 cross the top edge, W3 and W4 the outer edge, W5 and W6 the bottom edge. W1, W3 and W5 are 5 mm tape; W2, W4 and W6 are 12 mm.




## 2 Links at the edge and inboard



**E1 At the edge**
solder bridge across the wrapped edges

H1 · no net

**I1 10 mm inboard**
stranded wire, 30 mm loop

**I2 10 mm inboard**
one ribbon conductor, 30 mm loop

H2 · no net

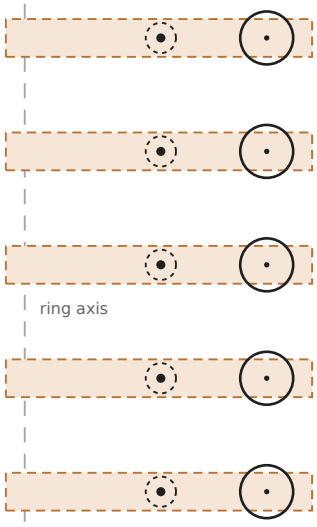
**E2 At the edge**
stranded wire, 30 mm service loop

**E3 At the edge**
one ribbon conductor, 30 mm loop

H3 · no net

## 3 The spine contract

15 mm pitch · 5 mm tape · 10 mm clear gap



**C1 GND**

**C2 BUS A**

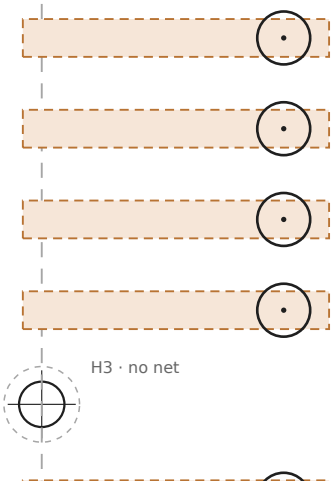
**C3 V+**

ring axis

**C4 BUS B**

**C5 GND**

## 4 Links on the ring axis



**X1 On the axis**
stranded wire laid 15 mm along it

**X2 On the axis**
one ribbon conductor laid along it

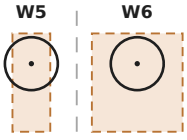
**X3 On the axis**
wire under a stitched card patch

**X4 On the axis**
solder bridge

**X5 Your version**

Print three cards: pages 1, 2 and 3 of a test book. Side A, this side, takes each link from the page before; side B sends it on to the page after.

With the book open, the back of one page lies beside the front of the next, so both ends of a link can be probed at once. Sheet 3 says how to build and test the links.



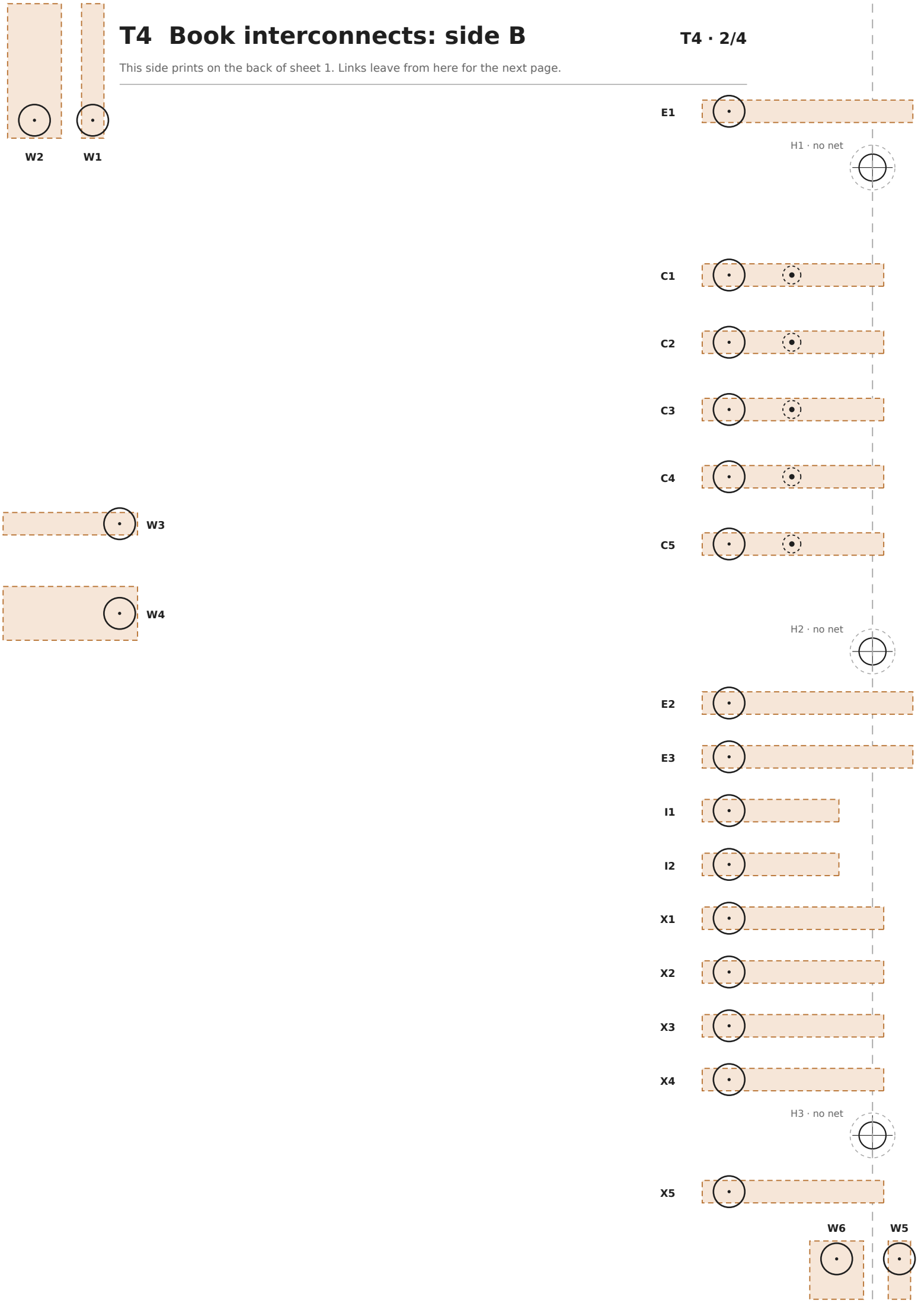



Print at 100 %: this bar should measure 100 mm.

# T4  Book interconnects: side B



This side prints on the back of sheet 1. Links leave from here for the next page.



# T4 Running the book interconnect tests T4 · 3/4

Print sheets 3 and 4 once on plain paper. Print sheets 1 and 2 double-sided on card, three times.

H1 · no net

## You need

- Three cards printed double-sided (sheets 1 and 2), a three-hole punch and a ring binder
- 5 mm and 12 mm copper tape, solder and a soldering iron
- Stranded hookup wire, about 28 AWG, and a strip of ribbon cable
- Thin card, clear tape, a needle and thread for strain relief
- A 100 Ω resistor (½ W or more), a 3 × AA holder and clip leads
- A multimeter with a 200 mV range

## 1 · Wraps

**1.** Lay W1 to W6 on side A of one card, run the tape over the edge and on down side B to the test point behind. Burnish the fold.

**2.** Measure each wrap from its side A test point to its side B test point, as in step 3.

**3.** Bend each wrapped edge back and forth ten times and measure again. Look for cracks.

## 2 · Build the test book

**1.** On all three cards, lay every link lane (E1 to X5) and the contract lanes (C1 to C5) on both sides. Keep all five contract lanes separate: do not join C1 to C5 at the contract block. Keep the clear gap between adjacent lanes at least twice as wide as the tape itself.

**2.** Make the contract block's wire vias: push a cut-off resistor leg through each via mark and solder it on both sides. Each via is an independent route between sides A and B.

**3.** Bind the cards in order, page 1 on top.

H2 · no net

**4.** For every row, join the back of page 1 to the front of page 2, and the back of page 2 to the front of page 3, the way the row's label says. Keep each solder joint at least 5 mm from the ring axis, except X4's, which tests a joint on it. On the axis rows, lay the wire or ribbon along the axis between the two joints.

**5.** Join each of C1 to C5 to its matching pin on the next page with a five-conductor strip of ribbon laid along the axis. Do not bridge C1 and C5 in the contract footprint. Mark the C1 edge with a stripe and keep that stripe on C1 at both ends; reversing the ribbon swaps BUS A and BUS B even though V+ and GND stay safe.

## 3 · Measuring a link

**1.** Clip the holder, the 100 Ω resistor and the link in a loop: holder +, resistor, the link's test point on the back of one page, the link, its test point on the front of the next page, holder −.

**2.** Measure the voltage across the resistor and divide by 100 Ω: that's the current, about 45 mA.

**3.** On the 200 mV range, measure from one test point to the other. 4.5 mV at 45 mA is 0.1 Ω.

## 4 · The spine contract chain

**1.** Temporarily join C1 and C5 with clip leads on page 1, outside the contract footprint. Clip the holder to page 1: red to C3 (V+), black to that temporary GND junction.

**2.** Temporarily join C1 and C5 on page 3. Clip the 100 Ω resistor across C3 and that temporary GND junction.

**3.** Measure C3 from one page to the next. Its drop divided by the total current must be 0.1 Ω or less.

**4.** Measure between the temporary GND junctions from one page to the next. The drop divided by total current must be 0.05 Ω or less. Do not calculate either GND conductor from total current; the current splits.

**5.** Remove the temporary GND joins. Test C1, C2, C4 and C5 one at a time in the separate loop from section 3; each must be 0.1 Ω or less. Confirm that the ribbon's striped edge is on C1 at both ends before applying power.

## 5 · Flex cycles

**1.** One cycle: open the spread between pages 1 and 2 all the way, close it, then do the same for pages 2 and 3.

**2.** Measure every link after 10, 100, 500 and 1,000 cycles.

**3.** While turning the pages slowly, watch the meter across the load on page 3. Any jump is a flicker; note which link moved.

**4.** Check every joint with a magnifier for cracks, lifted tape and torn paper.

## 6 · Repair

**1.** Unclip the holder. Cut every link to page 2, take it out, bind a fresh page 2 and resolder.

**2.** Note how long it took, then measure page 2's links again.

## Pass or fail

- **Pass:** each individual link and contract conductor is 0.1 Ω or less, the temporarily paired GND conductors are 0.05 Ω or less, and the load stays steady while the pages turn.
- **Fail:** a crack, lifted tape, torn paper or any flicker.
- Links at the edge move about 24 mm each cycle. If they fail, the V2 standard will prohibit them.

H3 · no net

Print at 100 %: this bar should measure 100 mm.

# T4  Results



Readings in mV at the measured current. Divide by the current for ohms.

### Wraps, side A to side B

| | New | Bent | Cracks? |
|---|---|---|---|
| **W1** | | | |
| **W2** | | | |
| **W3** | | | |
| **W4** | | | |
| **W5** | | | |
| **W6** | | | |

### The spine contract chain

Current, from the load: _____ mA

| | 1 to 2 | 2 to 3 |
|---|---|---|
| **C1  GND 1** | | |
| **C2  BUS A** | | |
| **C3  V+** | | |
| **C4  BUS B** | | |
| **C5  GND 2** | | |
| **C1 + C5  GND pair** | | |

### Results for the standard

| | |
|---|---|
| **Best link construction** | |
| **Pin-1 stripe stayed on C1 at every joint** | |
| **Closest safe distance from the axis** | |
| **Service loop needed** | |
| **Wrap tape: 5 mm or 12 mm** | |
| **Flex-life target met (cycles)** | |
| **Constructions to prohibit** | |

### Links, page 1 to page 2

| | 0 | 10 | 100 | 500 | 1,000 | Flicker? |
|---|---|---|---|---|---|---|
| **E1** | | | | | | |
| **E2** | | | | | | |
| **E3** | | | | | | |
| **I1** | | | | | | |
| **I2** | | | | | | |
| **X1** | | | | | | |
| **X2** | | | | | | |
| **X3** | | | | | | |
| **X4** | | | | | | |
| **X5** | | | | | | |

### Links, page 2 to page 3

| | 0 | 10 | 100 | 500 | 1,000 | Flicker? |
|---|---|---|---|---|---|---|
| **E1** | | | | | | |
| **E2** | | | | | | |
| **E3** | | | | | | |
| **I1** | | | | | | |
| **I2** | | | | | | |
| **X1** | | | | | | |
| **X2** | | | | | | |
| **X3** | | | | | | |
| **X4** | | | | | | |
| **X5** | | | | | | |